Programming Fundamentals

LAB-05 Submission

Name: **Rowan Aziz**

Roll No: **20**

Program: **Computer Science**

Section: **(E)**

Submitted to: **Sir Asad Majeed**

Date: **24/10/2026**

LAB Task-01:

Question: Write a C++ program to print Table of any number.

Handwritten Code:

Task 1: write a C++ Program to Print Table of any number.

```
#include<iostream>
using namespace std;
int main(){
    int num;
    cout<<"Enter any number:";
    cin>>num;
    for(int i=1; i<=10; i++)
    {
    cout<<num<<"*"<<i<<
    "="<<num*i<<endl;
    }
    return 0;
}
```

Output:

```
Enter any number: 5
5*1=5
5*2=10
5*3=15
5*4=20
5*5=25
5*6=30
5*7=35
5*8=40
5*9=45
5*10=50
```

Laptop Run Code:

Coding C++

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RUN MENU

```cpp

#include<iostream>
using namespace std;

int main()
{
    int num;

    cout << "Enter any number: ";
    cin >> num;

    for(int i = 1; i <= 10; i++)
    {
        cout << num << " * " << i << " = " << num * i << endl;
    }

    return 0;
}

```

OUTPUT:

Compile Result

```
Enter any number: 5
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50

[Process completed - press Enter]
```

LAB Task-02

Handwritten Code:

Task 2: C++ Programe to find Power of any number using for loop.

```
#include<iostream>
using namespace std;
int main() {
    int base, Power, result = 1;
    cout << "Enter base: ";
    cin >> base;
    cout << "Enter Power: ";
    cin >> Power;
  for (int i = 1;  i <= Power;  i++)
    {
      result = result * base;
    }
      cout << "result = " << result;
    return 0;
}
```

Output:

```
Enter base: 2
Enter Power: 3
result = 8
```

LAPTOP RUN CODE:

```cpp

#include<iostream>
using namespace std;

int main()
{
    int base, power, result = 1;

    cout << "Enter base: ";
    cin >> base;

    cout << "Enter power: ";
    cin >> power;

    for(int i = 1; i <= power; i++)
    {
        result = result * base;
    }

    cout << "Result = " << result;

    return 0;
}
```

OUTPUT:

```
Compile Result

Enter base: 2
Enter power: 3
Result = 8
[Process completed - press Enter]
```

LAB Task-03

Handwritten Code:

Task-3: Program to find Sum of Natural numbers Using Loop.

```
#include<iostream>
using namespace std;
  int main(){
      int n, Sum=0;
      cout<<"Enter a number:";
      cin>>n;
    for(int i=1; i<=n; i++)
     {
        Sum = Sum+1;
     }
        cout<<"Sum = "<<Sum;
    return 0;
  }
```

Laptop Run Code:

```cpp
#include<iostream>
using namespace std;

int main()
{
    int n, sum = 0;

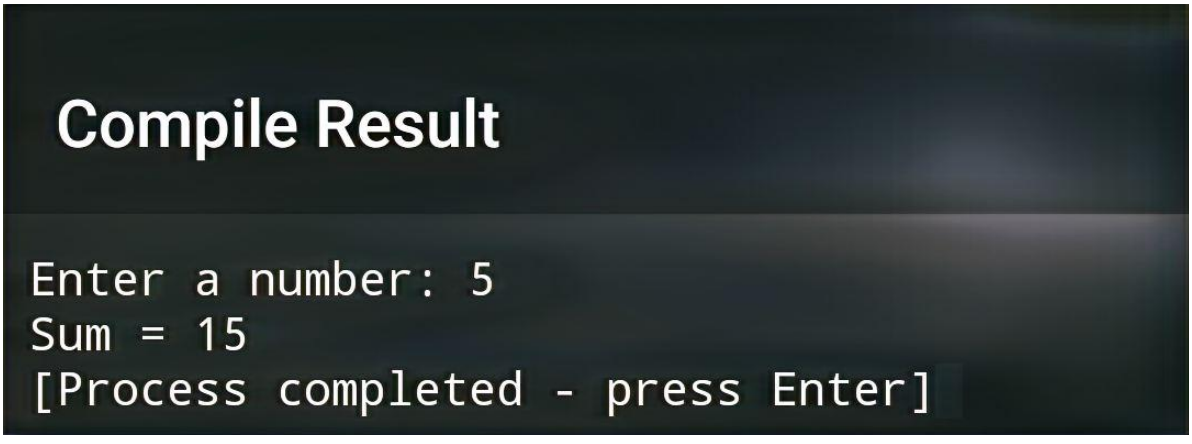
    cout << "Enter a number: ";
    cin >> n;

    for(int i = 1; i <= n; i++)
    {
        sum = sum + i;
    }

    cout << "Sum = " << sum;

    return 0;
}
```

OUTPUT:



LAB Task-04

Handwritten Code:

Task-4: C++ Program to find Sum of digits of a Number.

```
#include<iostream>
using namespace std;
int main() {
    int num, digit, sum =0;
    cout << "Enter a number: ";
    cin >> num;
    while (num >0)
    {
        digit = num % 10;
        Sum = sum+digit;
        num = num / 10;
    }
        cout << "Sum of digits = "<< Sum;
    return 0;
}
```

Output:

```
Enter a number : 350
Sum of digits = 8
```

Laptop Run Code:

```cpp

#include<iostream>
using namespace std;

int main()
{
    int num, digit, sum = 0;

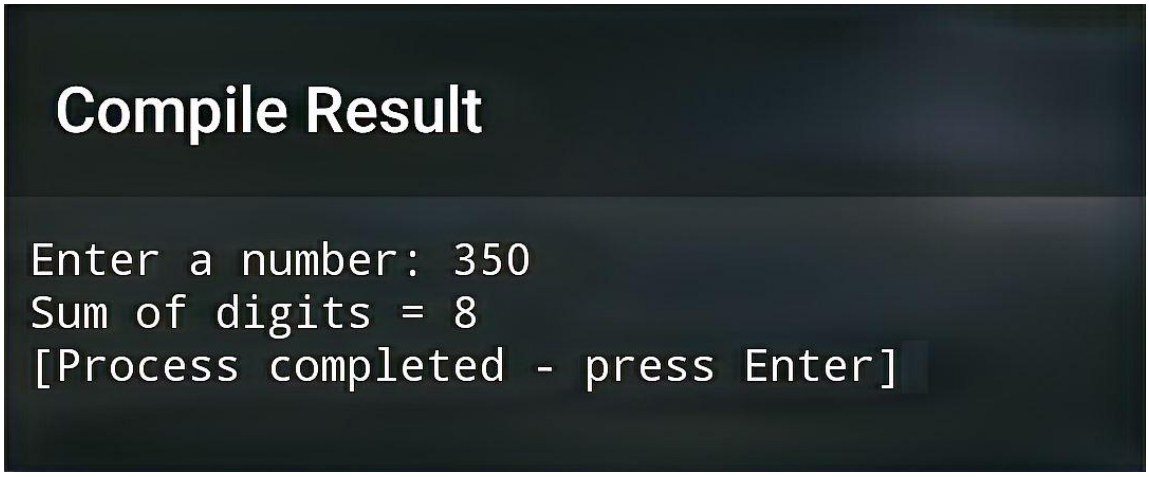
    cout << "Enter a number: ";
    cin >> num;

    while(num > 0)
    {
        digit = num % 10;
        sum = sum + digit;
        num = num / 10;
    }

    cout << "Sum of digits = " << sum;

    return 0;
}
```

UTPUT:



LAB Task-05

Handwritten Code:

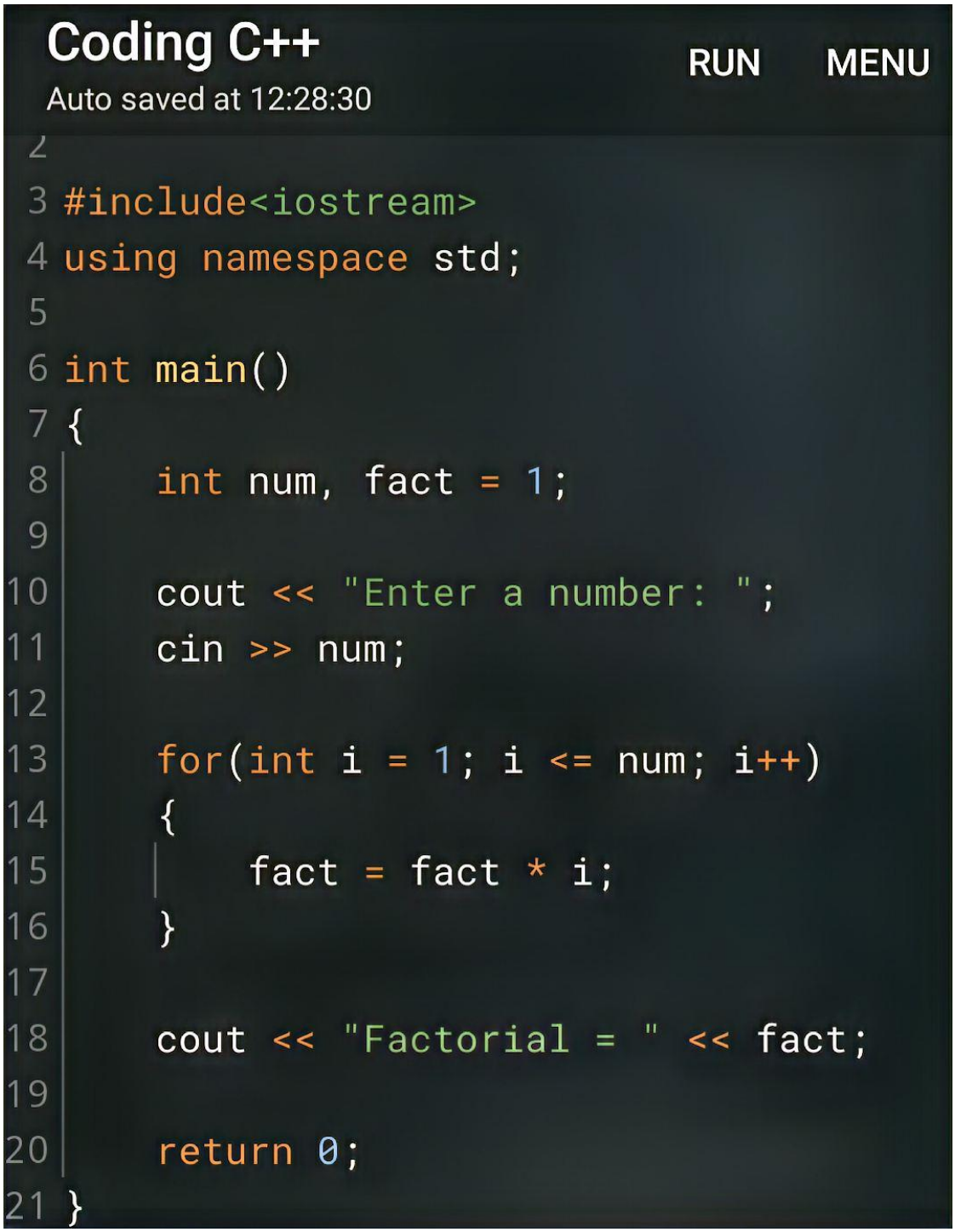
Task-5: C++ Program to find Factorial of a number.

```
# include <iostream>
using namespace std;
int main () {
int num, fact = 1;
cout << "Enter a number: ";
  cin >> num;
for (int i =1; i<=num; i++)
{
   fact = fact * i;
}
   cout << "Factorial = " << fact;
return 0;
}
```

output:

```
Enter a number: 4
Factorial = 24
```

Laptop Run Code:




```
#include<iostream>
using namespace std;

int main()
{
    int num, fact = 1;

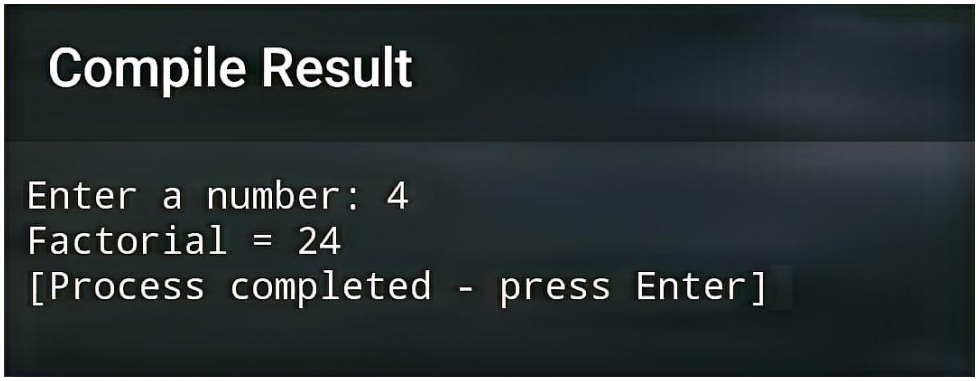
    cout << "Enter a number: ";
    cin >> num;

    for(int i = 1; i <= num; i++)
    {
        fact = fact * i;
    }

    cout << "Factorial = " << fact;

    return 0;
}
```

OUTPUT:



LAB Task-06

Handwritten Code:

Task-6: C++ Program to Print the Pattern:

```
*
**
***
****
*****
******
*******
```

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 7; i++)
    {
        for (int j = 1; j <= i; j++)
        {
            cout << "*";
        }
    }
    return 0;
}
```

Output:

```
*
**
***
****
*****
******
*******
```

Laptop Run Code:

Coding C++
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RUN MENU

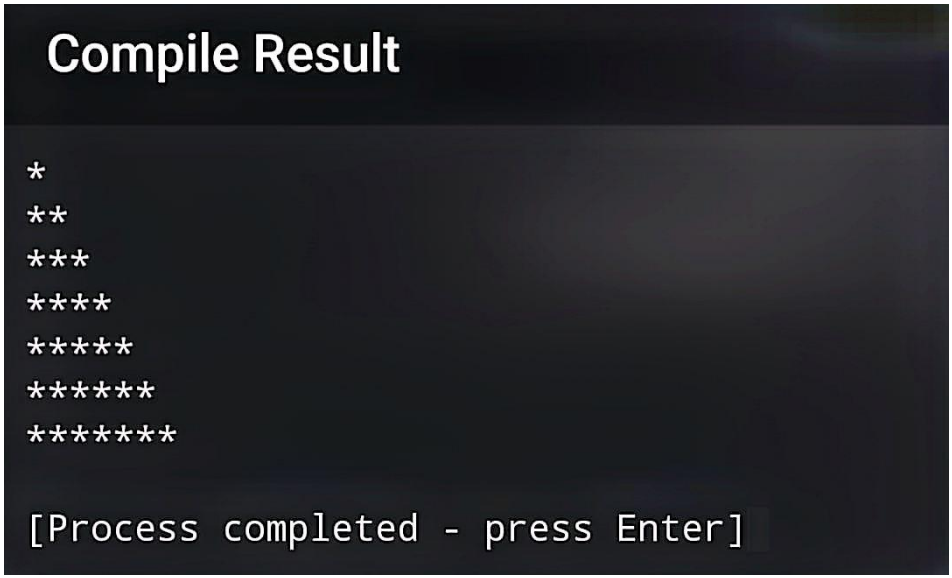
```cpp

#include<iostream>
using namespace std;

int main()
{
    for(int i = 1; i <= 7; i++)
    {
        for(int j = 1; j <= i; j++)
        {
            cout << "*";
        }
        cout << endl;
    }

    return 0;
}
```

OUTPUT:



Handwritten Code:

Lab 5

Task 7:

C++ Program to Print Total Number of Days in a Month Using Switch Case.

Code:

```
#include<iostream>
using namespace std;

int main()
{
   int month;

   cout << "Enter month number (1-12):";
   cin >> month;

   switch(month)
   {
      case 1:
      case 3:
      case 5:
      case 7:
      case 8:
      case 10:
      case 12:
            cout << "31 days";
            break;

      case 4:
      case 6:
      case 9:
      case 11:
            cout << "30 days";
            break;

      case 2:
            cout << "28 or 29 days";
            break;

      default:
            cout << "Invalid month number";
   }
   return 0;
}
```

Output:

```
Enter month number (1-12): 4
30 days
```

Explanation:

- The user enters a month number from 1 to 12.
- The switch statement checks the month number and prints the correct number of days.
- If the month number is 1, 3, 5, 7, 8, 10 or 12, it shows "31 days".
- If the month number is 4, 6, 9 or 11, it shows "30 days".
- If the month number is 2, it shows "28 or 29 days".
- If the month number is not valid, it shows "Invalid month number".

LAPTOP RUN CODE:

Coding C++

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RUN MENU

```cpp
#include<iostream>
using namespace std;

int main()
{
    int month;

    cout << "Enter month number (1-12): ";
    cin >> month;

    switch(month)
    {
        case 1:
        case 3:
        case 5:
        case 7:
        case 8:
        case 10:
        case 12:
            cout << "31 days";
            break;

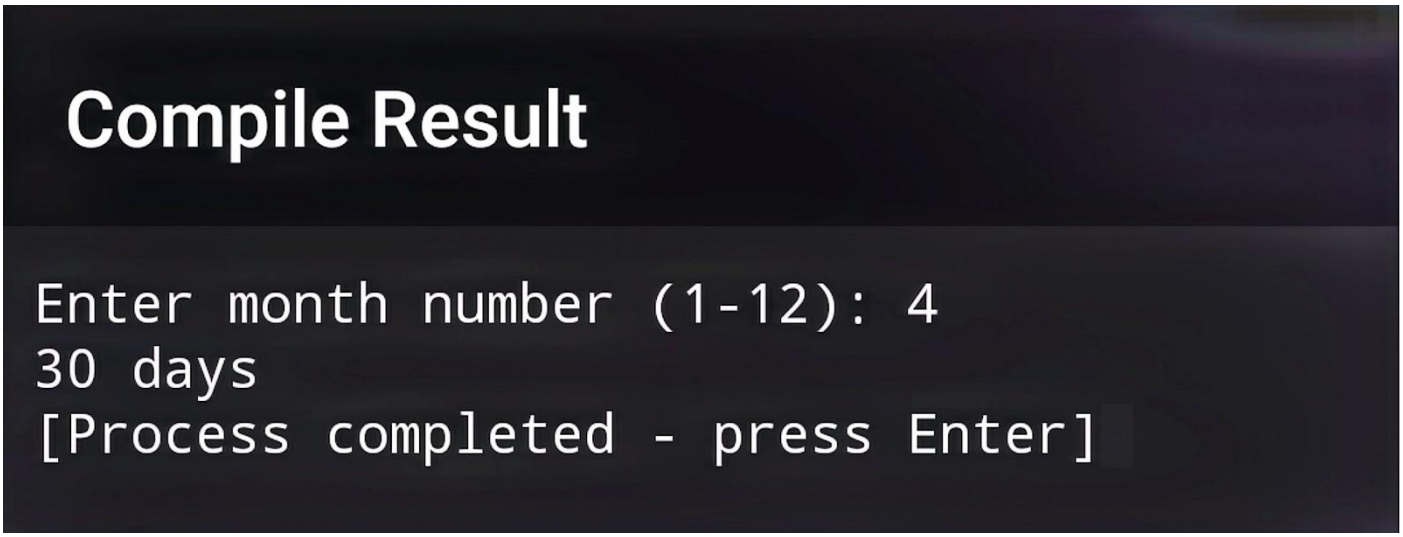
        case 4:
        case 6:
        case 9:
        case 11:
            cout << "30 days";
            break;

        case 2:
            cout << "28 or 29 days";
            break;

        default:
            cout << "Invalid month number";
    }

    return 0;
}
```

OUTPUT:



The End

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